

SWAPAN GUPTA CHOLLATI

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SUMMARY

A seasoned Data Engineer with over 4 years of experience specializing in building and optimizing data pipelines, particularly in the healthcare industry. My expertise spans across cloud platforms like Azure and AWS, where I've led projects that significantly enhanced data processing speeds, ensured data security, and streamlined ETL processes. I'm passionate about turning complex data into actionable insights and have a strong track record of improving query performance and enabling real-time analytics. I thrive in collaborative environments, always aiming to deliver solutions that align with business goals and drive impactful results.

EDUCATION

University of Maryland Baltimore County, Baltimore, USA Master of Science in Data Science	May 2023
Guru Nanak Institutions Technical Campus, Hyderabad Bachelor of Engineering in Electronics and Communication Engineering	Sept 2020

PROFESSIONAL EXPERIENCE

CVS Health <i>Data Engineer</i>	Remote, USA <i>Jun 2023 – Present</i>
- Built and optimized ETL pipelines using Apache Beam (Dataflow) and Cloud Composer (Airflow), reducing manual efforts by 40%. Ensured data integrity and accuracy through robust validation and quality checks across ETL pipelines. - Enhanced BigQuery performance by implementing partitioning, clustering, and materialized views, improving query execution speed by 50%. - Designed and implemented automated ETL processes for electronic health records (EHR) and patient data using Python and SQL, boosting data retrieval speeds by 35%. - Designed a Google Cloud Storage (GCS)-based data lake for seamless ingestion of structured and semi-structured data from Cloud Pub/Sub, Cloud Storage, and Cloud SQL, cutting data latency from hours to seconds. - To ensure HIPAA compliance, implemented IAM policies, VPC Service Controls, and Cloud DLP (Data Loss Prevention) for PHI/PII security. Optimized BigQuery and storage costs, reducing expenses by 40% and 30%, respectively, through query tuning and lifecycle management. - Integrated diverse healthcare data sources, including HL7, FHIR APIs, flat files, and databases, into a Snowflake data warehouse. - Automated CI/CD pipelines using Terraform, Cloud Build, and GitHub Actions, reducing ETL deployment time by 60%. Collaborating with Data Scientists, Analysts, and DevOps teams, Enabled seamless data access via BigQuery ML, Looker, and Cloud Functions. - Designed and deployed serverless APIs using Cloud Functions and Cloud Run to expose data securely for downstream applications. Integrated API Gateway with IAM roles to control data access and ensure compliance with enterprise security standards. - Applied industry best practices in version control, branching strategies, and CI/CD workflows. - Developed automated reporting solutions using SQL and Python, providing real-time insights for healthcare stakeholders.	
Amazon <i>Data Engineer</i>	Bengaluru, India <i>Oct 2020 Jun 2021</i>
- Designed and deployed real-time data streaming pipelines using Amazon Kinesis, Kafka, and AWS Lambda, reducing data processing latency from hours to seconds for operational reporting. - Implemented real-time analytics with AWS Kinesis Data Analytics and Redshift Streaming, enabling instant data insights and improving business decision-making speed by 40%. - Developed event-driven architectures using AWS SNS, SQS, and Lambda, ensuring seamless data flow and reducing processing bottlenecks in distributed systems. - Deployed AWS EMR and Spark for big data processing, enabling the management of large-scale healthcare datasets and performing complex data transformations and aggregations. - Built a real-time fraud detection system leveraging Amazon Kinesis, AWS Glue, and Redshift, enabling instant anomaly detection and reducing fraud-related losses. - Integrated real-time healthcare data ingestion pipelines using FHIR APIs, Amazon Kinesis, and EMR, ensuring patient data accessibility. - Optimized IoT real-time data processing using AWS IoT Core, Kinesis, and DynamoDB, improving data throughput and storage efficiency for millions of sensor events.	

- Designed and implemented a change data capture (CDC) pipeline with AWS DMS and Kinesis, enabling real-time database replication with minimal downtime.
- Developed real-time monitoring and alerting using AWS CloudWatch, Lambda, and SNS, improving system reliability and reducing incident response time by 50%.
- Developed custom scripts in Python and SQL for data extraction, transformation, and analysis, automating routine data tasks and increasing efficiency by 35%.

Sigma InfoSolutions

Junior Data Engineer

Hyderabad, India

Sept 2019 – Sept 2020

- Developed and maintained real-time ETL pipelines using Apache Kafka, AWS Kinesis, and Spark Streaming, ensuring seamless data ingestion from multiple sources while maintaining data integrity and consistency.
- Designed and deployed interactive real-time dashboards in Tableau and Power BI, integrating live data from AWS Kinesis, Redshift, and BigQuery, enabling stakeholders to make data-driven decisions in real-time.
- Collaborated with cross-functional teams to define real-time data processing requirements and built event-driven architectures using AWS Lambda, SNS, and SQS, ensuring low-latency data availability.
- Implemented real-time data warehouses and data lakes using Amazon Redshift, Google BigQuery, and AWS S3, supporting instant analytics and streaming data storage for business intelligence.
- Enhanced data quality and validation by deploying real-time anomaly detection models using AWS Glue, Spark Streaming, and ML models in SageMaker, improving data reliability and accuracy.
- Optimized SQL queries and Python scripts for real-time data pipelines, leveraging window functions and stream joins in Apache Flink and Kinesis Analytics, improving data processing speed by 30%.
- Monitored and maintained real-time data pipelines, implementing CloudWatch, Datadog, and Prometheus for real-time logging, alerting, and issue resolution, reducing downtime by 50%.
- Integrated relational (MySQL) and NoSQL (MongoDB, DynamoDB) databases with real-time data streams, improving data availability and enhancing cross-system interoperability.

TECHNICAL SKILLS

Methodologies: SDLC, Agile/ Scrum, Waterfall

Language & Databases: Python, SQL, R, SCALA, MySQL, MS SQL Server, ETL.

Python Packages: Pandas, NumPy, Matplotlib, SciPy, Scikit-Learn, SeaBorn, PyTorch, ggplot2, Plotly

Data Components: HDFS, Hue, MapReduce, PIG, Hive, HCatalog, HBase, Sqoop, Impala, Zookeeper, Flume, Kafka, Yarn, Cloudera Manager, Kerberos, Pyspark Airflow, Kafka Snowflake

Data Analytics Skills: Data Manipulation, Data Cleaning, Data Visualization, Exploratory Data Analysis, Data Analysis

Others: AWS, AZURE(Databricks), NLP, A/B Testing, Hypothesis testing, ETL, Hadoop, Spark, Big Query, Apache Airflow,

Tools: Tableau, Power BI, Advanced Excel, Visual Studio, GIT, Jupyter Notebook

Version Control: Git, GitHub **Operating Systems:** Windows, macOS

PROJECTS

Big Data Processing and Feature Extraction | [GitHub](#)

- Managed and reviewed an extensive dataset of 9 million images from Google Open Image Dataset V5 using Python (Pandas, NumPy) and EDA techniques, achieving a key feature discovery for enhanced classifier efficiency through MobileNet model application.
- Pioneered a custom convolutional and sequential layer model, attaining an exceptional confidence level of 3.10, validating the model's proficiency in alternative feature extraction and image pattern discernment.

Global GDP Dynamics Analysis: Uncovering Economic Indicators through Data Analytics | [GitHub](#)

- Examined World Bank data on World Development Explorer using Python (Pandas, Seaborn, Plotly), unveiling pivotal GDP factors through expert Exploratory Data Analysis (EDA), identifying top 6 nations with the highest GDP.
- Executed comprehensive data organization and integrity checks, leveraging Python and Matplotlib, to isolate key economic growth indicators, enhancing understanding of global economic dynamics.